



Optimal excitation trajectories for mechanical systems identification[☆]

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ABSTRACT

System identification for multibody mechanical systems such as robots and vehicles is typically formulated as a parameter optimization problem, in which the model parameters are chosen to minimize the difference between measured and predicted trajectories of the system. The problem is made difficult by the large number of parameters of different scales and physical units, and also noisy and incomplete measurements. Even more critically, the choice of reference trajectory has a decisive impact on the accuracy and robustness of the identification procedure. In this paper we propose a set of geometric optimal excitation criteria that can be optimized to generate high-quality reference trajectories. The resulting optimal trajectories are coordinate- and frame-invariant, and can be obtained efficiently and robustly using recursive analytic gradients of the criteria. For high-dimensional systems that can execute only a limited range of feasible trajectories, we also show how our geometrical framework can be used to optimally identify a reduced set of parameters for the given set of trajectories. The improved robustness and accuracy of our geometric approach vis-à-vis existing methods is demonstrated through both numerical and hardware experiments involving robot manipulators and a high-dimensional humanoid robot.

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1. Introduction

From industrial robots and humanoids to ground and aerial vehicles, mechanical systems constitute a large and important class of input-output systems that are the subject of model-based control. In their most common manifestation, the equations of motion for controlled mechanical systems are second-order differential equations of the form

$$\tau = M(q, \Phi)\ddot{q} + b(q, \dot{q}, \Phi), \quad (1)$$

where $(q, \dot{q}) \in \mathbb{R}^n \times \mathbb{R}^n$ is the state, $\tau \in \mathbb{R}^n$ is the control input, $\Phi \in \mathbb{R}^N$ are the model parameters, $M(q, \Phi) \in \mathbb{R}^{n \times n}$ is the symmetric positive-definite mass matrix, and $b(q, \dot{q}, \Phi) \in \mathbb{R}^n$ contains potential, Coriolis, and other dynamics terms. For mechanical systems comprised of coupled rigid bodies, the model

parameters appear linearly in the dynamics, so that the equations of motion can also be written in the form (Atkeson, An, & Hollerbach, 1986)

$$\tau = \Gamma_B(q, \dot{q}, \ddot{q})\Phi_B, \quad (2)$$

with respect to the structurally identifiable set of base parameters $\Phi_B \in \mathbb{R}^{N_B}$ ($N_B < N$), where $\Phi_B = B\Phi$ for some constant full-rank matrix $B \in \mathbb{R}^{N_B \times N}$.

System identification typically proceeds by first taking measurements of $\tau, q, \dot{q}, \ddot{q}$ at multiple time instances $0 \leq t_1 \leq \dots \leq t_m \leq T$ along some reference trajectory, and solving the following least-squares optimization (Atkeson et al., 1986) for Φ_B :

$$\min_{\Phi_B} \|\mathbf{A}_B \Phi_B - \mathbf{y}\|_{\Sigma^{-1}}^2, \quad (3)$$

where

$$\mathbf{A}_B = [\Gamma_B^T|_{t_1}, \dots, \Gamma_B^T|_{t_m}]^T \in \mathbb{R}^{nm \times N_B}, \quad (4)$$

$$\mathbf{y} = [\tau(t_1)^T, \dots, \tau(t_m)^T]^T \in \mathbb{R}^{nm}, \quad (5)$$

the notation $\|\cdot\|_Q$ denotes the vector norm with respect to the symmetric positive-definite matrix Q , and $\Sigma \in \mathbb{R}^{nm \times nm}$ is the sample covariance matrix associated with the observation $\mathbf{y}$.

Extending from the linear least squares method, various robust identification techniques have been proposed which have proven to be robust against more practical conditions on the measurement noise as well as modeling errors (Gautier, Janot,

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& Vandanjon, 2008; Janot, Vandanjon, & Gautier, 2009, 2014; Olsen, Swevers, & Verdonck, 2002; Ramdani & Poignet, 2005). Meanwhile, it is to note that there is still a fundamental limitation on the unbiased estimation performance known as the Cramér–Rao bound; even under perhaps the most mild condition (no modeling error with the regressor matrix $\mathbf{A}_B$ being noise-free and the measurement $\mathbf{y}$ being Gaussian), the variance of any unbiased estimator $\hat{\Phi}_B^*$ of Φ_B is lower-bounded by,

$$\text{var}[\hat{\Phi}_B^*] \geq (\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)^{-1}. \quad (6)$$

In practice, $\mathbf{A}_B$ is more often than not ill-conditioned, meaning that only some reduced set, or linear combination of the parameters Φ_B , are effectively identifiable. Further complicating matters is that the Φ_B cannot assume any arbitrary vector value, but rather must satisfy a set of complex nonlinear constraints dictated by physics, e.g., masses cannot be negative, and inertia tensors must be symmetric positive-definite (Sousa & Cortesão, 2014; Wensing, Kim, & Slotine, 2018). Φ_B also typically lumps together parameters with different physical units whose values may differ by orders of magnitude. All of these complications make it difficult to assess the accuracy of the parameters in a physically meaningful way, and are quite often the source of numerical instability in the estimation procedure.

Referring to (6), the achievable estimation variance being directly dependent on the regressor matrix $\mathbf{A}_B$ implies that the choice of reference trajectory samples can significantly impact the performance of the identification. Loosely speaking, the reference trajectory samples should be sufficiently rich so as to excite the full spectrum of the dynamics, which can be high-dimensional and quite complex; the number of parameters N_B can range from 30–60 for a typical industrial manipulator to several hundred for a humanoid robot. Further complicating matters is that for such high-dimensional systems, the available measurements may be limited by the number and location of the sensors (e.g., only a subset of a robot's joints may be equipped with torque sensors), and the set of admissible reference trajectories may be restricted by stability, safety, and other considerations (e.g., a vehicle should avoid roll-over, a humanoid robot should always maintain postural stability).

In our previous work Lee, Kwon, and Park (2018), Lee and Park (2018) and Lee, Wensing, and Park (2019), we have demonstrated that many of these challenges can be mitigated by appealing to coordinate-invariant, differential geometric perspective on the mass-inertial parameter space. More specifically, in Lee et al. (2019), we have proposed regularized least square formulation in which the original linear least squares objective (3) is appended with the geometric distance from the nominal mass-inertial parameter values; many of the difficulties in regularized mass-inertial parameter identification for multibody mechanical systems can be traced to the coordinate-dependent way in which the distance between two different mass distributions is measured (typically the Euclidean distance between the parameter values) and that there exists a more natural, coordinate-invariant way that takes into account the manifold structure of the mass-inertial parameters. However, we note that while the regularized least squares method can offer robustness and significant increase in estimation precision, it is still advisable to collect sufficiently rich trajectory samples at first hand for the purpose of accurate identification for the biased nature of the regularized estimates.

Taking the same coordinate-invariant, geometric perspective as our point of departure, in this paper we consider the problem of generating sufficiently rich trajectories for mechanical systems identification. More specifically, we address the problem of how to effectively generate reference trajectory samples that encode maximal information—the optimal excitation trajectory generation problem—taking into account physical constraints and measurement characteristics, thus leading to more accurate and robust parameter identification.

1.1. Related works

Nearly all of the existing approaches (Armstrong, 1989; Ayusawa, Rioux, Yoshida, Venture, & Gautier, 2017; Bonnet et al., 2016; Gautier & Khalil, 1992; Swevers, Ganseman, Bilgin, De Schutter, & Van Brussel, 1997; Wilson, Schultz, & Murphey, 2014) to optimal excitation trajectory generation for multibody mechanical systems have been based on using classical optimal experimental design criteria that are directly associated with the parameter estimation variance (Donev, Atkinson, & Tobias, 2007; Moakher & Batchelor, 2006; Pukelsheim, 2006; Schwabe, 2012). Other types of criteria can be chosen to, e.g., minimize the prediction variance of specific outputs of interest (Pukelsheim, 2006). Further, more recent directions which have gained special interest in the control systems field include closed-loop experiment design, robust experiment design, and also the idea of “identification for control” in which the performance of the model-based controller is directly maximized over the input space (Gevers, 2005; Jansson & Hjalmarsson, 2005; Pronzato, 2008; Rojas, Welsh, Goodwin, & Feuer, 2007). In this paper, we confine our interest to the most standard case of generating excitation trajectories for offline parameter identification. The problem is typically formulated as minimizing some symmetric function f of the eigenvalues $\lambda = (\lambda_1, \dots, \lambda_N)$ of the Fisher information matrix $\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B$ (or, equivalently, the inverse of the covariance matrix of the least-squares estimator) evaluated over the trajectory samples, i.e.,

$$\min_{q(t), t \in [0, T]} f(\lambda(\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)) \quad (7)$$

It is important to note that the criteria described in this way are generally not invariant with respect to the choice of coordinates.¹ Coordinate choices involving general multibody mechanical systems include the positions and orientations of reference frames attached to the bodies comprising the system, as well as the linear reparametrization of the observable parameters through the matrix $\mathbf{B}$. Clearly a meaningful measure of information contained in trajectories should not depend on these coordinate choices.

From a more pragmatic perspective, it has been pointed out in Presse and Gautier (1993) that the various coordinate-dependent criteria proposed in the literature are often ill-defined in a way that leads to an imbalanced distribution of information over the parameters, producing a set of trajectory samples from which the estimation of certain parameters remains highly sensitive to measurement noise. Several approaches that are essentially scale normalization techniques for the numerical values of the Fisher information matrix, e.g., Dette (1997), Donev et al. (2007), López-Fidalgo, Rivas-López, and Fernández-Garzón (2007), Morgan and Wang (2010) and Presse and Gautier (1993), have been shown to alleviate this sensitivity somewhat. However, to the best of our knowledge, general coordinate invariance of the excitation criteria for mechanical systems identification has yet to be addressed in a rigorous manner or validated with systematic and thorough experiments.

1.2. Contributions

In this paper we present a geometric, coordinate-free framework in which to formulate optimal trajectory excitation criteria. Our main focus will be on a criterion in which the information matrix is weighted with the Riemannian metric evaluated at the nominal value, with symmetric functions f of its eigenvalues admitting physically meaningful interpretations while preserving coordinate-invariance. The proposed coordinate-free framework frees the designer to focus solely on the intrinsic aspects of the

¹ One notable exception is the D-optimality criterion, $f(\lambda) = \lambda_1 \cdots \lambda_N$.

problem, e.g., choosing a physically meaningful Riemannian metric and symmetric function f . Moreover, structural similarities between our proposed criterion and existing criteria allows for the use of the same numerical optimization algorithms for, e.g., evaluating the gradient of the criterion (Ayusawa et al., 2017), by performing simple linear transformations of the variables via the chain rule.

Finally, we show how the same geometric framework can be applied to infer the optimal reduced set of effectively identifiable parameters from the given trajectory samples. Such reduction of the model parameters can be useful not only for finding optimal excitation trajectories, but also parameter identification for complex high-dimensional systems like humanoid robots, in which the execution of reference trajectories that sufficiently excite the entire structurally identifiable set of parameters is difficult or impractical to achieve.

2. Preliminaries

In this section, we revisit various conventional optimality criteria for excitation trajectories used for mechanical systems identification. We also discuss through both examples and analysis how the choice of reference frames, base parameter set, and physical units explicitly enters into formulations of existing criteria. This discussion serves to motivate a new set of coordinate-invariant criteria to be proposed in the next section.

2.1. Excitation criteria for mechanical systems

Mechanical systems identification is typically reduced to a parameter estimation problem on linear observation models. Specifically, for an n -link open chain robot with joint variables $\mathbf{q} \in \mathbb{R}^n$ and joint torque vector $\boldsymbol{\tau} \in \mathbb{R}^n$, the dynamic equations are of the form

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q}, \dot{\mathbf{q}})\ddot{\mathbf{q}} + \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) = \boldsymbol{\Gamma}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\Phi}. \quad (8)$$

Here $\boldsymbol{\Phi} = [\phi_1^T, \dots, \phi_n^T]^T \in \mathbb{R}^{10n}$ denotes the complete set of inertial parameters for the n links, with the inertial parameter vector $\phi_i \in \mathbb{R}^{10}$ for link i commonly expressed in the form

$$\phi_i = [m_i, \mathbf{h}_i, I_i^{xx}, I_i^{yy}, I_i^{zz}, I_i^{xy}, I_i^{yz}, I_i^{zx}] \in \mathbb{R}^{10}, \quad (9)$$

where the scalar m_i is the mass, $\mathbf{h}_i = m_i \cdot \mathbf{p}_i \in \mathbb{R}^3$ is the first moment with $\mathbf{p}_i \in \mathbb{R}^3$ the position of the mass center, and $\mathbf{I}_i \in \mathbb{R}^{3 \times 3}$ is the symmetric 3 by 3 rotational inertia matrix, all represented with respect to a body-fixed reference frame. More generally, other dynamic terms like frictions or elastic force can also be typically factored in a linear form. For instance, joint Coulomb force on each joint is commonly modeled as, $\tau_f^i(q_i, \dot{q}_i, k_c^i) = \text{sgn}(\dot{q}_i)k_c^i$ (c.f., (76)). However, unless otherwise noted, we only consider the complete set of mass-inertial parameters for parameter estimation and excitation. We also note that the model vector $\boldsymbol{\Phi}$ being linear with respect to the control inputs $\boldsymbol{\tau}$ or *inverse dynamics model* is a fairly general assumption for multibody mechanical systems, including ground and aerial vehicles (Venture, Ripert, Khalil, Gautier, & Bodson, 2006), humanoid robots (Ayusawa, Venture, & Nakamura, 2014; Bonnet et al., 2016), etc.

Now, consider the linear regression model

$$\boldsymbol{\tau} = \boldsymbol{\Gamma}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\Phi} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon) \quad (10)$$

where the observation $\boldsymbol{\tau}$ is subject to additive Gaussian noise with covariance $\boldsymbol{\Sigma}_\epsilon \in \mathbb{R}^{n \times n}$. We assume that observations are taken at m time instances $t_1, \dots, t_m$ along some reference trajectory $\mathbf{q}(t)$. Then, the least-squares estimation problem for $\boldsymbol{\Phi}$ is formulated as

$$\min_{\boldsymbol{\Phi}} \|\mathbf{A}\boldsymbol{\Phi} - \mathbf{y}\|_{\boldsymbol{\Sigma}^{-1}}^2. \quad (11)$$

where $\boldsymbol{\Sigma} = \text{diag}(\boldsymbol{\Sigma}_\epsilon, \dots, \boldsymbol{\Sigma}_\epsilon) \in \mathbb{R}^{mn \times mn}$ is a block diagonal matrix of $\boldsymbol{\Sigma}_\epsilon$, and

$$\mathbf{A} = [\boldsymbol{\Gamma}(t_1)^T, \dots, \boldsymbol{\Gamma}(t_m)^T]^T \in \mathbb{R}^{nm \times 10n}, \quad (12)$$

$$\mathbf{y} = [\boldsymbol{\tau}(t_1)^T, \dots, \boldsymbol{\tau}(t_m)^T]^T \in \mathbb{R}^{nm}. \quad (13)$$

As alluded to earlier, one of the unique features of the multibody dynamic equations (8) is that the regressor matrix $\boldsymbol{\Gamma}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ is inherently rank-deficient. More precisely, the dynamic equations can be factored into a more compact form

$$\boldsymbol{\Gamma}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\Phi} \equiv \boldsymbol{\Gamma}_B(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\Phi}_B, \quad (14)$$

so that only reduced linear combinations of $\boldsymbol{\Phi}$ of the form

$$\boldsymbol{\Phi}_B = \mathbf{B}\boldsymbol{\Phi} \in \mathbb{R}^{n_B}, \quad (n_B \leq 10n) \quad (15)$$

(the *base parameters*) are identifiable. This restriction is in fact due to the kinematic structure of the coupled rigid body system (Atkeson et al., 1986; Wensing, Niemeyer, & Slotine, 2017).

Remark 1. The representation of the base parameters is not unique, as different choices of basis vectors can be used, e.g., $\boldsymbol{\Phi}_B = \mathbf{S}\boldsymbol{\Phi}_B = (\mathbf{S}\mathbf{B})\boldsymbol{\Phi} = \hat{\mathbf{B}}\boldsymbol{\Phi}$ and $\boldsymbol{\Gamma}_B = \boldsymbol{\Gamma}_B\mathbf{S}^{-1}$, with $\mathbf{S} \in \mathbb{R}^{n_B \times n_B}$ any nonsingular square matrix.

Taking the reduced least-square formulation (3)–(5), the unique solution is given in closed-form as

$$\boldsymbol{\Phi}_B^* = (\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B)^{-1} \mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{y}, \quad (16)$$

Since the observation $\mathbf{y}$ is a random vector following the Gaussian distribution $p(\mathbf{y}|\boldsymbol{\Phi}_B) = \mathcal{N}(\mathbf{A}_B \boldsymbol{\Phi}_B, \boldsymbol{\Sigma})$, the least-square estimator $\boldsymbol{\Phi}_B^*$ (16) is also a Gaussian random vector with mean and covariance matrix explicitly given by $\mathbb{E}[\boldsymbol{\Phi}_B^*] = \boldsymbol{\Phi}_B$ and

$$\text{var}[\boldsymbol{\Phi}_B^*] = (\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B)^{-1}. \quad (17)$$

It can be observed that the least-square estimator $\boldsymbol{\Phi}_B^*$ is an unbiased estimator of $\boldsymbol{\Phi}_B$. Moreover, the estimator is efficient in the sense that the estimator covariance matrix achieves the Cramer–Rao lower bound, i.e.,

$$\text{var}[\boldsymbol{\Phi}_B^*] = \mathbf{M}_B^{-1}, \quad (18)$$

where $\mathbf{M}_B$ is the Fisher information matrix given by

$$\mathbf{M}_B = \mathbb{E}[\nabla_{\boldsymbol{\Phi}_B} \log p(\mathbf{y}|\boldsymbol{\Phi}_B) \cdot \nabla_{\boldsymbol{\Phi}_B} \log p(\mathbf{y}|\boldsymbol{\Phi}_B)^T] \quad (19)$$

$$= \mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B. \quad (20)$$

Observe that both the information matrix (20) and the estimator covariance matrix (17) depend explicitly on the trajectory samples $\{\mathbf{q}(t_i), \dot{\mathbf{q}}(t_i), \ddot{\mathbf{q}}(t_i)\}_{i=1}^m$. Sampling the trajectory so as to maximize the information or, equivalently, to minimize the least-squares estimator variance, can be regarded as a problem in *optimal experiment design* (Pukelsheim, 2006). Optimality criteria are typically defined as symmetric functions $f: \mathbb{R}_+^n \rightarrow \mathbb{R}$ of the eigenvalues $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_n)$ of the Fisher information matrix (or the inverse of estimator covariance matrix), i.e.,

$$J = f(\boldsymbol{\lambda}(\mathbf{M}_B)) \quad (21)$$

$$= f(\boldsymbol{\lambda}(\text{var}[\boldsymbol{\Phi}_B^*]^{-1})) \quad (22)$$

$$= f(\boldsymbol{\lambda}(\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B)). \quad (23)$$

Some of the most popular widely used criteria are

- A-optimality: $f(\boldsymbol{\lambda}) = \sum_{i=1}^n 1/\lambda_i = \text{tr}((\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B)^{-1})$
- D-optimality: $f(\boldsymbol{\lambda}) = -\sum_{i=1}^n \log \lambda_i = -\log |\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B|$
- E-optimality: $f(\boldsymbol{\lambda}) = \lambda_{\max}(\mathbf{A}_B^T \boldsymbol{\Sigma}^{-1} \mathbf{A}_B)^{-1}$
- Condition number: $f(\boldsymbol{\lambda}) = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}} = \text{cond}(\boldsymbol{\Sigma}^{-1/2} \mathbf{A}_B)$

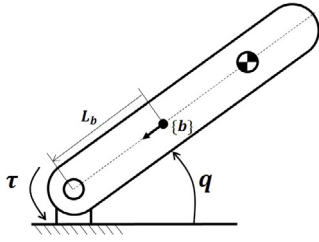


Fig. 1. 1-link, 1-dof robot.

Minimizing the above criteria typically correspond to estimates that minimize variance or maximize information. The exception is the condition number, which attempts to balance uncertainties across the estimation parameters, i.e., $\text{var}[\Phi_B^*] \simeq \alpha \mathbf{1}$ for some positive scalar α (here $\mathbf{1}$ denotes the identity matrix).

Various design criteria have been used in the context of optimal excitation trajectory generation for multibody system identification, e.g., the condition number (Armstrong, 1989; Gautier & Khalil, 1992; Swevers et al., 1997), D-optimality (Swevers et al., 1997), E-optimality (Armstrong, 1989; Wilson et al., 2014), etc. Numerical algorithms for generating optimal excitation trajectories are discussed in Section 4.

2.2. Coordinate invariance and normalization

Here we show explicitly how formulating a measure of information through criteria of the form (23) can be affected by the particular coordinates used to characterize the dynamic model. To illustrate, consider a simple 1-dof robot shown in Fig. 1. The dynamic equations are given by

$$\tau = \underbrace{\begin{bmatrix} L_b^2 \ddot{q} - g L_b \cos(q) & -2L_b \ddot{q} + g \cos(q) & \ddot{q} \end{bmatrix}}_{= \Gamma(q, \dot{q}, \ddot{q}) \in \mathbb{R}^{1 \times 3}} \underbrace{\begin{bmatrix} m \\ h_b \\ I_b \end{bmatrix}}_{= \Phi \in \mathbb{R}^3}, \quad (24)$$

where $L_b \in \mathbb{R}$ is the distance to the joint axis from the origin of the body-fixed frame $\{b\}$, and $m \in \mathbb{R}^+$, $h_b \in \mathbb{R}$, and $I_b \in \mathbb{R}^+$ are the mass, first, and second moments of the 1-d mass distribution of the link, respectively, all described with respect to the body-fixed frame $\{b\}$. More compactly,

$$\tau = \underbrace{\begin{bmatrix} -2L_b \ddot{q} + g \cos(q) & \ddot{q} \end{bmatrix}}_{= \Gamma_B(q, \dot{q}, \ddot{q}) \in \mathbb{R}^{1 \times 2}} \underbrace{\begin{bmatrix} h_b - mL_b \\ I_b - mL_b^2 \end{bmatrix}}_{= \Phi_B \in \mathbb{R}^2}, \quad (25)$$

where

$$\Phi_B = \underbrace{\begin{bmatrix} -L_b & 1 & 0 \\ -L_b^2 & 0 & 1 \end{bmatrix}}_{= B \in \mathbb{R}^{2 \times 3}} \Phi. \quad (26)$$

Consider a family of sinusoidal excitation trajectories of the form

$$q(t; A, q_0) = A \sin(2\pi t) + q_0, \quad 0 \leq t \leq 2, \quad (27)$$

parametrized by the amplitude A and offset q_0 . The infinite-dimensional optimization problem (7) can then be reduced to a two-dimensional parameter optimization problem on $(A, q_0) \in \mathbb{R}^2$. Fig. 2 shows the contour map of the A-optimality criterion (which is to be minimized for richer/lower information/variance). It can be observed that just by relocating the position of the body-fixed frame $\{b\}$ or changing physical units, the shape of the

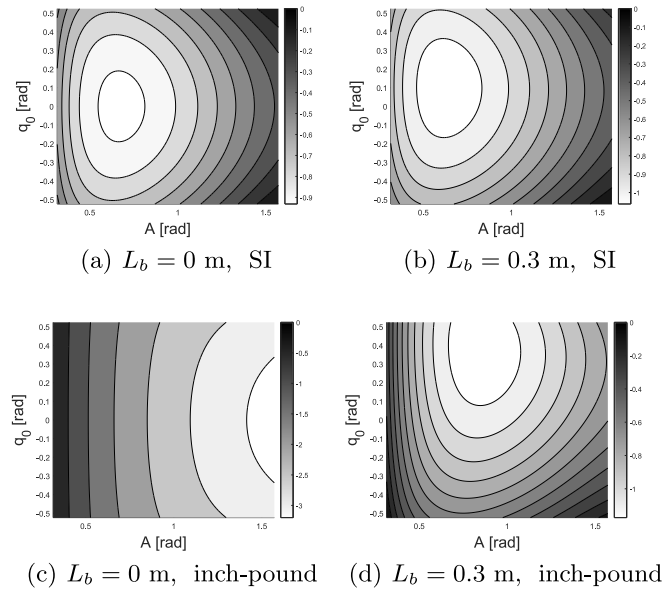


Fig. 2. Contour maps of A-optimality excitation criterion for the one-dof robot, shown at logarithmic scale and evaluated with different coordinate choices.

objective function changes dramatically, resulting in substantially different optimal excitation trajectories.

More generally, coordinate choices for describing the multibody dynamic model (8) include not only reference frames and physical units of measurement, but also linear reparametrizations of the base parameter vector Φ_B . To illustrate, let B_1 and B_2 be two arbitrary choices of matrix B . As long as the identifiable set of parameters is consistent without regard to its representation, there should exist some nonsingular square matrix $S_{12} \in \mathbb{R}^{N_B \times N_B}$ such that $B_1 = S_{12} B_2$ holds. The Fisher information matrix M_B then exhibits a linear transformation with respect to the change of base parameter representation, i.e.,

$$M_{B_2} = S_{12}^T M_{B_1} S_{12}. \quad (28)$$

Since the set of eigenvalues is not invariant to such linear transformations of the Fisher information matrix, i.e.,

$$\lambda(M_{B_1}) \neq \lambda(S_{12}^T M_{B_1} S_{12}), \quad (29)$$

the choice of base parameter representation also affects conventional experimental designs (23).

In the work of Presse and Gautier (1993), which is most relevant to ours, although the issue of coordinate invariance is not explicitly raised, it is pointed out that standard ways of specifying a multibody dynamic model (with respect to a particular coordinate system) can more often than not lead to poorly scaled values of the model parameters. This affects the characterization of the excitation criteria in a way that makes certain parameters overly sensitive to measurement noise. To circumvent this problem, a normalized form of the Fisher information matrix is proposed so as to balance the contribution of each entry in M_B . The proposed criterion is of the form

$$J = f(\lambda(M_B \cdot [\text{diag}(\Phi_B^0)]^2)) \quad (30)$$

$$= f(\lambda(\text{var}[[\text{diag}(\Phi_B^0)]^{-1} \cdot \Phi_B^*]^{-1})) \quad (31)$$

$$= f(\lambda(A_B^T \Sigma^{-1} A_B \cdot [\text{diag}(\Phi_B^0)]^2)), \quad (32)$$

where Φ_B^0 is some nominal parameter vector, which for mechanical systems can be easily obtained from, e.g., CAD models.

While this can alleviate the aforementioned difficulties to some extent, we claim that by appealing to geometric methods,

normalization can in fact be achieved in a much more physically meaningful way while also ensuring coordinate invariance of the criteria.

3. Geometric normalized excitation criteria

In this section we propose a new normalized optimal excitation criterion given nominal values of the parameters. The criterion is concisely formulated to be coordinate-invariant, while the relative sensitivities or variabilities of the estimation parameters are quantified in a geometrically natural manner. As a result, many of the aforementioned difficulties of existing coordinate-based criteria can be mitigated.

3.1. Motivation

The general experimental design problem as reviewed in the previous section is an intrinsically multivariate problem (Pukelsheim, 2006): the matrix-valued information of the estimation parameter vector is reduced to a scalar-valued criterion to be optimized, with the traditional criterion as formulated in (23) one possible solution. We instead construct a scalar representation of this matrix-valued information from a geometric coordinate-invariant perspective.

The information matrix given by the inverse of the parameter covariance matrix, i.e., $\text{var}[\Phi_B^*]^{-1} = A_B^T \Sigma^{-1} A_B$, can be viewed as a matrix-valued quantity that represents the local variabilities of the estimated parameter Φ_B^* accruing from the observation noise. Defining a scalar representation of the information matrix can be thought of as quantifying some representative scalar-valued variability of the estimation parameter Φ_B^* on the parameter space. Toward this end, it is noted that any geometric quantifications, e.g., local displacement or volume element, require a well-defined metric predefined on the space.

3.2. Affine-invariant metric on base parameters

In this section, we define a natural metric on the space of base parameters $\Phi_B = B\Phi$. To do so we first present a physically meaningful metric on the space of full inertial parameters Φ , then demonstrate how this metric can be naturally projected to the base parameter space under the surjective linear mapping B .

Recently it has been shown in Wensing et al. (2018) that the physically consistent inertial parameters ϕ of a single rigid body can be identified under the one-to-one linear mapping $P: \mathbb{R}^{10} \rightarrow S(4)$ with the requirement that the 4×4 symmetric matrix

$$P = P(\phi) = \begin{bmatrix} \frac{1}{2}\text{tr}(I) \cdot \mathbb{1} - I & h \\ h^T & m \end{bmatrix} \in S(4) \quad (33)$$

be positive-definite, i.e., $P(\phi) \succ 0$. Such a 4×4 symmetric matrix representation of the inertial parameters, P_b , has also been referred to as pseudo-inertia matrix in the literature Jazar (2007). Owing to the explicit one-to-one (invertible) linear mapping P , the space of physically consistent inertial parameters of a single rigid body inherits the smooth manifold structure of the space of 4×4 symmetric positive-definite matrices, $\mathcal{P}(4)$ (Moakher & Batchelor, 2006). Also, the following affine-invariant Riemannian metric (Moakher & Batchelor, 2006) on the space of physically consistent inertial parameters can be defined (Lee & Park, 2018):

$$ds^2 = \frac{1}{2} \text{tr} (P(\phi)^{-1} dP(\phi))^2 = d\phi^T g(\phi) d\phi. \quad (34)$$

The coordinate representation of the metric $g(\cdot) \in \mathbb{R}^{10 \times 10}$ is defined over the values of ϕ making $P(\phi)$ a positive-definite matrix.

We now show that this metric is invariant to the choice of coordinates. Given a symmetric matrix representation $P \in \mathcal{P}(4)$

of the mass-inertia properties of a rigid body, any change of reference frame or change in physical units/scales can be expressed by a transformation $P \mapsto GPG^T$ for some $G \in GL(4)$, where $GL(4)$ represents the general linear group of 4×4 nonsingular matrices (Lee & Park, 2018). This operation represents a $GL(4)$ group action $*$ on $\mathcal{P}(4)$ defined by

$$G * P = GPG^T, \quad (35)$$

and since the affine-invariant metric is invariant to this group action (Lee & Park, 2018), frame and scale/unit invariance of the proposed metric is guaranteed.

Remark 2. The proposed metric also captures more naturally the distance between the underlying mass distributions of two rigid bodies compared to the standard Euclidean metric, i.e., $ds^2 = d\phi^T d\phi$. We refer the reader to Lee and Park (2018) and Lee et al. (2019) for further geometric features of the metric.

For an n -link multibody system, the Riemannian metric on the space $\mathcal{M}$ of complete set of inertial parameters can be defined as a direct summation, i.e.,

$$ds^2 = \sum_{i=1}^n d\phi_i^T g(\phi_i) d\phi_i = d\Phi^T G(\Phi) d\Phi, \quad (36)$$

$$G(\Phi) = \text{diag}(g(\phi_1), \dots, g(\phi_n)) \in \mathbb{R}^{10n \times 10n}. \quad (37)$$

With this definition, denote by

$$G_0 \triangleq G(\Phi_0) \quad (38)$$

the Riemannian metric evaluated at some given nominal inertial parameter value Φ_0 . We note that G_0 can also serve as a well-defined coordinated-invariant Riemannian metric on $\mathcal{M}$.² Further, flat Riemannian manifold $\mathcal{M}$ admitting the global coordinate representation Φ with constant Riemannian metric G_0 can be characterized as a vector space equipped with an inner product,

$$\langle x, y \rangle_{\mathcal{M}} = x^T G_0 y, \quad (39)$$

where $x, y \in \mathbb{R}^{10n}$ are formally the coordinate values of the elements in the tangent space of $\mathcal{M}$ which is isomorphic to $\mathcal{M} \simeq \mathbb{R}^{10n}$.³ Assuming that the region of interest on $\mathcal{M}$ is near the nominal value Φ_0 , the flat Riemannian manifold structure with metric G_0 allows for a reasonable approximation of the curved Riemannian structure with Riemannian metric $G(\Phi)$.

Remark 3. The main reason why we avoid using the original parameter-dependent Riemannian metric $G(\Phi)$ is to construct the excitation criterion (presented in Section 3.4) to be independent of the unknown true parameter Φ . It is noted however that experimental designs for nonlinear observation models generally possess similar difficulties from the criterion, or to be more precise the Fisher information matrix, being dependent on the unknown true parameters (Pronzato, 2008; Rojas et al., 2007).

² It is sufficient to demonstrate the coordinate invariance of the constant metric for $n = 1$ given by $ds^2 = \frac{1}{2} \text{tr} ((P(\phi_0)^{-1} dP(\phi))^2)$. Using the coordinate transformation rule expressed by the group action $*$ (35), i.e., $P(\phi_0) \leftarrow GP(\phi_0)G^T$ and $dP(\phi) \leftarrow GdP(\phi)G^T$, straightforward calculation reveals the coordinate invariance of the metric.

³ Technically, in order to formally characterize such $\mathcal{M}$ as the vector space, the global coordinates Φ should now span the entire $\mathbb{R}^{10n}$ space, not the open convex subset in $\mathbb{R}^{10n}$ previously enforced with the physical consistency condition, $P(\phi_i) \succ 0$ for $i = 1, \dots, n$. However, for the metric G_0 to provide a positive-definite inner product, the nominal value Φ_0 is still required to satisfy the physical consistency condition.

Now, a natural consequence for endowing a metric on the base parameter space $\mathcal{N}$ would be to enforce isometry with respect to $\mathcal{M}$, i.e.,

$$d\Phi_B^T H_0 d\Phi_B \doteq d\Phi G_0 d\Phi, \quad (40)$$

where $H_0 \in \mathbb{R}^{N_B \times N_B}$ is a constant Riemannian metric on $\mathcal{N}$ to be defined. If the mapping B is not just surjective but one-to-one, i.e., B is a full-rank square matrix, the isometry is trivially achieved by letting $H = B^{-T} G B^{-1}$. For a general surjective mapping B , the displacement $d\Phi_B$ on $\mathcal{N}$ actually imposes a redundancy on $d\Phi$ with the relation $Bd\Phi = d\Phi_B$. The following minimal norm solution can be chosen to define a unique choice of metric on $\mathcal{N}$:

$$\begin{aligned} d\Phi_B^T H_0 d\Phi_B &\doteq \min_{d\Phi \in \{x: Bx = d\Phi_B\}} d\Phi G_0 d\Phi \\ &= d\Phi_B^T (B G_0^{-1} B^T)^{-1} d\Phi_B. \end{aligned} \quad (41)$$

Consequently, the constant Riemannian metric H_0 , pushforwarded from $\mathcal{M}$ to $\mathcal{N}$, is given by

$$H_0 \triangleq (B G_0^{-1} B^T)^{-1}. \quad (42)$$

We use the following coordinate-free inner product notation on $\mathcal{N}$:

$$\langle x, y \rangle_{\mathcal{N}} \triangleq x^T H_0 y, \quad (43)$$

where $x, y \in \mathbb{R}^{N_B}$ are formally the coordinate values of the elements in the tangent space of $\mathcal{N}$. Unless otherwise noted, the abbreviated notation $\langle \cdot, \cdot \rangle$ would be used for $\langle \cdot, \cdot \rangle_{\mathcal{N}}$.

Remark 4. The inherited metric $(B G_0^{-1} B^T)^{-1}$ in (42) has the following alternative interpretation. Let the prior distribution of Φ be given such that the precision matrix is proportional to the natural metric, i.e., $\text{var}[\Phi]^{-1} = \gamma \cdot G_0$ for some constant positive scalar value γ (the covariance matrix inverse $\text{var}[\cdot]^{-1}$ corresponding to the information matrix also represents the precision of the estimate). Then the precision matrix of the prior base parameter $\Phi_B = B\Phi$ is straightforwardly given by

$$\begin{aligned} \text{var}[\Phi_B]^{-1} &= \text{var}[B\Phi]^{-1} \\ &= (B \text{var}[\Phi] B^T)^{-1} \\ &= \gamma \cdot (B G_0^{-1} B^T)^{-1}, \end{aligned}$$

which turns out to be proportional to the metric in (42) derived from solving (41).

3.3. Fisher information matrix as pullback metric

In this section, we suggest the Fisher information matrix Φ_B as another choice of constant Riemannian metric on the base parameter space $\mathcal{N}$. As opposed to the metric H_0 being pushforwarded from $\mathcal{M}$, now we show that $M_B = A_B^T \Sigma^{-1} A_B$ is a metric pullbacked from the observation space $\mathcal{O}$.

First of all, one natural choice of metric on the observation space $\mathcal{O}$ can be given as the inverse of the measurement noise covariance matrix Σ^{-1} ; a small variation of the observation vector dy can be quantified as,

$$ds^2 = dy^T \Sigma^{-1} dy. \quad (44)$$

When the linear mapping $A_B : \mathcal{N} \rightarrow \mathcal{O}$ is an immersion (i.e., A_B is full rank), a small variation $d\Phi_B$ of the base parameter vector can be measured from the metric (44) as,

$$ds^2 = dy^T \Sigma^{-1} dy \quad (45)$$

$$= (A_B d\Phi_B)^T \Sigma^{-1} (A_B d\Phi_B) \quad (46)$$

$$= d\Phi_B^T (A_B^T \Sigma^{-1} A_B) d\Phi_B. \quad (47)$$

Hence, the Fisher information matrix M_B explicitly given as $M_B = A_B^T \Sigma^{-1} A_B$ can be interpreted as the pullback⁴ of the metric Σ^{-1} defined on the observation space $\mathcal{O}$ under the linear regressor mapping $A_B : \mathcal{N} \rightarrow \mathcal{O}$.

3.4. Geometric normalized criterion

So far, we have presented both geometric and statistical interpretations of the Fisher information matrix M_B and the nominal constant Riemannian metric H_0 . From a geometric perspective, both M_B and H_0 are metric tensors of $\mathcal{N}$ that are pushforwarded from $\mathcal{M}$ and pullbacked from $\mathcal{O}$ respectively. From a statistical perspective, M_B and H_0 are inverse of the linear least squares estimation covariance matrix and the scaled prior covariance matrix (see Remark 4) respectively.

Recall that the Fisher information matrix M_B admits the explicit coordinate transformation rule given by (28). Mathematically, such a symmetric matrix-type variable admitting the coordinate transformation rule of the form (28) is characterized as a symmetric 2-covariant tensor. Riemannian metric and inverse of covariance matrix are also examples of symmetric 2-covariant tensors.

Meanwhile, it is evident from (29) that the eigenvalues are generally not the invariants of symmetric 2-covariant tensors. Instead, the eigenvalues relative to the metric tensor defines a class of coordinate-invariant measures of the given symmetric 2-covariant tensors (Eells & Sampson, 1964). More specifically, given the symmetric 2-covariant tensor M on the Riemannian manifold $\mathcal{M}$ with Riemannian metric G , the roots of the characteristics equation, $\det(\lambda G - M)$, or equivalently the eigenvalues of $M G^{-1}$, are coordinate invariant. This leads to the following coordinate-invariant formulation of the excitation criterion.

Proposition 5 (Geometric Normalized Criterion). Symmetric functions f of the eigenvalues of the normalized information matrix

$$J = f(\lambda(A_B^T \Sigma^{-1} A_B \cdot H_0^{-1})), \quad (48)$$

or equivalently,

$$J = f(\lambda_{\neq 0}(A^T \Sigma^{-1} A \cdot G_0^{-1})), \quad (49)$$

where $A = A_B B$, are invariant with respect to the choice of base parameter representations, body-fixed reference frames, and physical units (here $\lambda_{\neq 0}(X)$ denotes the nonzero eigenvalues of the symmetric positive-semidefinite matrix X).

Proof of Proposition 5 is given in Appendix. Referring to Fig. 3, the generalized eigenvalues of the pullback metric $A_B^T \Sigma^{-1} A_B$ relative to the natural metric H_0 can be understood as the intrinsic measure that capture how much the local geometry gets distorted from the mapping A_B in comparison to the natural geometry predefined on the parameter space $\mathcal{N}$ with metric H_0 . Alternatively, viewing the fisher information matrix as the inverse of the estimation covariance matrix,

$$\begin{aligned} J &= f(\lambda(\text{var}[\Phi_B^*]^{-1} H_0^{-1})) \\ &= f\left(\lambda\left(H_0^{-1/2} \text{var}[\Phi_B^*]^{-1} H_0^{-1/2}\right)\right) \\ &= f\left(\lambda\left(\text{var}\left[H_0^{1/2} \Phi_B^*\right]^{-1}\right)\right), \end{aligned} \quad (50)$$

⁴ Let $\mathcal{M}$ be a Riemannian manifold with metric g and a mapping $f : \mathcal{N} \rightarrow \mathcal{M}$ be an immersion. Coordinate representation of the pullback metric tensor f^*g of the metric g via f is given as $J^T G J$ where G and J are the coordinate representation of the metric g and the Jacobian of the mapping f respectively.

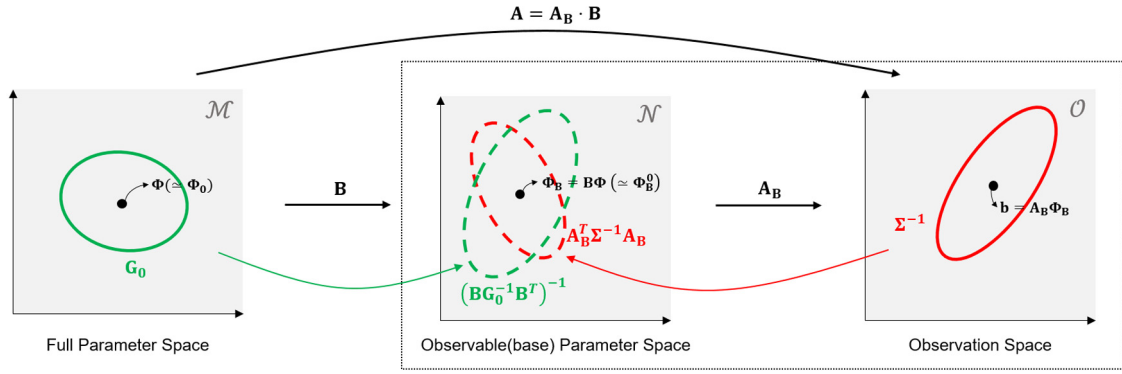


Fig. 3. Geometric distortion view of the proposed excitation criterion. The metrics characterized with (green and red) bold ellipses are predefined respectively on the full parameter space and observation space. The (green and red) dashed ellipses represent respectively the pushforward and pullback of the metrics to the observable (base) parameter space under the linear mappings $\mathbf{B}$ and $\mathbf{A}_B$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the proposed family of coordinate-invariant optimality criteria can be interpreted as normalized version of the traditional optimality criteria. More specifically, the traditional optimality criterion exactly matches to our criterion when using the orthonormal basis representation of the parameters, i.e., $\Phi_B^* \leftarrow \mathbf{H}_0^{1/2} \Phi_B^*$.

Below is a more detailed, explicit comparison of the proposed criterion with their unnormalized counterparts.

- **A-optimality:** The traditional A-optimality criterion is defined by the expected parameter error using the standard Euclidean metric, i.e.,

$$\begin{aligned} \text{tr}((\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)^{-1}) &= \text{tr}(\text{var}[\Phi_B^*]) \\ &= \mathbb{E}[(\Phi_B^* - \Phi_B)^T (\Phi_B^* - \Phi_B)], \end{aligned}$$

In contrast, the proposed normalized A-optimality criteria is defined to be the expected quadratic parameter error but in coordinate-invariant form, i.e.,

$$\begin{aligned} \text{tr}((\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)^{-1} \mathbf{H}_0) & \\ &= \mathbb{E}[(\Phi_B^* - \Phi_B)^T \mathbf{H}_0 (\Phi_B^* - \Phi_B)]. \end{aligned} \quad (51)$$

- **D-optimality:** D-optimal designs are already invariant to reparametrizations of the parameters. Indeed, the normalized D-optimality criterion turns out to be the same as the original D-optimality criterion shifted by the constant value $\log |\mathbf{H}_0|$:

$$\begin{aligned} -\log |\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{H}_0^{-1}| & \\ &= -\log |\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B| + \log |\mathbf{H}_0| \end{aligned} \quad (52)$$

- **E-optimality:** The E-optimality criterion is defined by the maximum eigenvalue of the estimator covariance matrix:

$$\begin{aligned} \lambda_{\max}((\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)^{-1}) &= \lambda_{\max}(\text{var}[\Phi_B^*]) \\ &= \max_{\mathbf{x}^T \mathbf{x} = 1} \mathbf{x}^T \text{var}[\Phi_B^*] \mathbf{x} \\ &= \max_{\mathbf{x}^T \mathbf{x} = 1} \text{var}[\mathbf{x}^T \Phi_B^*], \end{aligned}$$

which as derived above can geometrically be understood as the largest variance of the estimated parameter projected along the unit-normalized directional vector $\mathbf{x}$. Similarly, the proposed normalized E-optimality criterion

$$\lambda_{\max}((\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B)^{-1} \mathbf{H}_0) \quad (53)$$

is equivalent to

$$\max_{\langle \mathbf{x}, \mathbf{x} \rangle = 1} \text{var}[\langle \mathbf{x}, \Phi_B^* \rangle], \quad (54)$$

where the Euclidean inner products are replaced by $\langle \cdot, \cdot \rangle$ defined in (43).

- **Condition number:** Traditional condition number optimization seeks to find the estimator with isotropic variance, i.e. $\text{var}[\Phi_B^*] \sim \mathbf{I}$, while the proposed normalized condition number

$$\sqrt{\frac{\lambda_{\max}(\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{H}_0^{-1})}{\lambda_{\min}(\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{H}_0^{-1})}} \quad (55)$$

is at its minimum when the precision of the estimator is proportional to the metric $\mathbf{H}_0$, i.e.,

$$\text{var}[\Phi_B^*]^{-1} = \mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \sim \mathbf{H}_0.$$

Remark 6. Bayesian experimental design (Chaloner, 1984) is a class of experimental design framework that explicitly take into account the prior distribution in both the experimental design and estimation procedures. When the prior distribution is given by $\Phi_B \sim \mathcal{N}(\Phi_B^0, \mathbf{R})$, the posterior distribution of Φ_B is also Gaussian with precision matrix given by $\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B + \mathbf{R}$. Then, the so-called Bayesian D-optimal design criterion is defined as,

$$-\log |\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B + \mathbf{R}^{-1}|. \quad (56)$$

When we restrict the prior precision matrix to be proportional to the natural metric, i.e., $\mathbf{R}^{-1} = \gamma \cdot \mathbf{H}_0$ for some positive scalar γ (see Remark 4), the Bayesian D-optimal design criterion can be represented in the form of (48) as

$$\begin{aligned} &-\log |\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B + \gamma \mathbf{H}_0| \\ &= -\log |\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{H}_0^{-1} + \gamma \mathbf{I}| - \log |\mathbf{H}_0| \\ &= f(\lambda(\mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{H}_0^{-1})) + \text{const.}, \end{aligned}$$

where,

$$f(\lambda) = -\sum_{i=1}^n \log(\lambda_i + \gamma). \quad (57)$$

Under our geometric framework, the normalization given in Presse and Gautier (1993) can be understood as adopting a diagonal scale-normalizing (but coordinate-dependent) metric $[\text{diag}(\Phi_B^0)]^{-2}$ on the parameter space (see Eq. (32)). While $[\text{diag}(\Phi_B^0)]^{-2}$ is clearly a scale-invariant metric, it is not frame-invariant. For instance, referring to the exemplar case of 1-dof robot presented in Section 2.2, if one places the origin of the coordinate frame on the nominal mass center, i.e., $L_b = h_b^0/m^0$, the first entry of Φ_B^0 becomes a zero value. Then, the scale-normalization on the first entry of Φ_B^* is no longer well-defined. Perhaps, for such simple models involving only a few number of parameters, a more careful choice of coordinate frames or

reparametrization may allow one to circumvent similar problems arising from the coordinate dependent nature of the criterion. However, finding a canonical parameter representation for that matter can become increasingly unintuitive and time-consuming for complex high-dimensional mechanical systems, as the base parameters for general coupled rigid body systems are typically complex and lump together parameters with different physical units whose values may differ by orders of magnitude.

We would argue that our approach offers a systematic way of quantifying the variability of the multi-scale, non-trivially correlated base parameters of multibody mechanical systems in a completely coordinate-free manner. Also, we would like to point out that the metric $\mathbf{H}_0$ (42) presented in the paper is just one particular choice. Other choices of metric may be more useful or appropriate for the particular usage of the model. This leads to the following final remark on some connections to the prediction error related criteria.

Remark 7. Prediction error related criteria (like I-, V-, L- optimality criteria Pukelsheim, 2006), unlike the parameter error related criteria discussed so far, usually represent prediction variance of a set of specific output variables as a known function of the parameters, rather than the parameters themselves. For instance, when one is only interested in predicting a fixed linear combinations of parameters, i.e., $\mathbf{z} = (z_1, \dots, z_r) = \mathbf{L} \Phi_{\mathbf{B}} \in \mathcal{Z} = \mathbb{R}^r$ with $\mathbf{L}$ some known constant $r \times n_{\mathbf{B}}$ matrix, the weighted L-optimality criteria is given by

$$J = \text{tr}((\mathbf{A}_{\mathbf{B}}^T \Sigma^{-1} \mathbf{A}_{\mathbf{B}})^{-1} \mathbf{L}^T \mathbf{A}^{-1} \mathbf{L}) \quad (58)$$

$$= \mathbb{E}[\|\mathbf{z}^* - \mathbf{z}\|_{\mathbf{A}^{-1}}^2],$$

which represents the prediction variance of the best linear unbiased estimator $\mathbf{z}^* = \mathbf{L} \Phi_{\mathbf{B}^*}$ of $\mathbf{z}$. Note the similarity between the proposed normalized A-optimality criterion (51) and (59). Interpreting the weighted L-optimality criterion within our geometric formulation, the original intrinsic metric choice $\mathbf{H}_0$ is now replaced with $\mathbf{L}^T \mathbf{A}^{-1} \mathbf{L}$ which is the pullback of the metric $\mathbf{A}^{-1}$ on the output prediction space $\mathcal{Z}$ under the mapping $\mathbf{L} : \mathcal{N} \rightarrow \mathcal{Z}$.

4. Excitation trajectory optimization

Finding optimal excitation trajectories is an infinite-dimensional variational problem, and in this section we describe direct numerical methods for finding finite-dimensional approximate solutions. For the conventional (i.e., non-geometric) criteria proposed in the literature, the variational formulation of the problem is of the form

$$\min_{q(t), t \in [0, T]} J = f(\lambda(\mathbf{M}_{\mathbf{B}})), \quad (59)$$

where $\mathbf{M}_{\mathbf{B}} = \mathbf{A}_{\mathbf{B}}^T \Sigma^{-1} \mathbf{A}_{\mathbf{B}} = \sum_{i=1}^m \Gamma_{\mathbf{B}}(t_i)^T \Sigma_{\epsilon}^{-1} \Gamma_{\mathbf{B}}(t_i)$, typically subject to dynamic equations of the form (8) and other physical constraints such as joint angle, velocity, and torque limits. Direct methods parametrize the trajectory $q(t)$ in terms of a finite set of parameters $\mathbf{P} = \{p_1, \dots, p_{n_p}\}$, e.g., B-spline control points. The parametrized trajectory $\mathbf{q}(t, \mathbf{P})$ and its time derivatives $\dot{\mathbf{q}}(t, \mathbf{P})$, $\ddot{\mathbf{q}}(t, \mathbf{P})$ are now explicit functions of $\mathbf{P}$, and the variational problem reduces to the following finite-dimensional parameter optimization:

$$\min_{\mathbf{P}} J = f \left(\lambda \left(\sum_{i=1}^m \Gamma_{\mathbf{B}}(t_i, \mathbf{P})^T \Sigma_{\epsilon}^{-1} \Gamma_{\mathbf{B}}(t_i, \mathbf{P}) \right) \right). \quad (60)$$

Note that the regressor $\Gamma_{\mathbf{B}}(\mathbf{q}(t_i), \dot{\mathbf{q}}(t_i), \ddot{\mathbf{q}}(t_i))$ is also an explicit function of $\mathbf{P}$.

Efficient and reliable numerical optimization of J in (60) requires analytic gradients with respect to the parameters $\mathbf{P}$; the

accumulation of numerical errors resulting from finite-difference estimates of the true gradient more often than not leads to ill-conditioning and poor convergence. Gradients of J can be analytically obtained via application of the chain rule (Ayusawa et al., 2017):

$$\frac{\partial J}{\partial \mathbf{p}} = \sum_{i=1}^{N_{\mathbf{B}}} \frac{\partial f}{\partial \lambda_i} \text{tr}((\nabla_{\mathbf{M}_{\mathbf{B}}} \lambda_i)^T \frac{\partial \mathbf{M}_{\mathbf{B}}}{\partial \mathbf{p}}), \quad (61)$$

where the notation $\nabla_{\mathbf{X}}$ denotes the derivative operator on a scalar function with respect to a matrix $\mathbf{X}$, i.e., (i, j) th component of a matrix $\nabla_{\mathbf{X}} f$ is defined as $\frac{\partial f}{\partial x_{ij}}$. The term $\frac{\partial \mathbf{M}_{\mathbf{B}}}{\partial \mathbf{p}}$ is given by

$$\frac{\partial \mathbf{M}_{\mathbf{B}}}{\partial \mathbf{p}} = \sum_{i=1}^m \left[\frac{\partial \Gamma_{\mathbf{B}}(t_i, \mathbf{P})^T}{\partial \mathbf{p}} \Sigma_{\epsilon}^{-1} \Gamma_{\mathbf{B}}(t_i, \mathbf{P}) + \Gamma_{\mathbf{B}}(t_i, \mathbf{P})^T \Sigma_{\epsilon}^{-1} \frac{\partial \Gamma_{\mathbf{B}}(t_i, \mathbf{P})}{\partial \mathbf{p}} \right], \quad (62)$$

where

$$\frac{\partial \Gamma_{\mathbf{B}}(t_i, \mathbf{P})}{\partial \mathbf{p}} = \frac{\partial \Gamma(t_i, \mathbf{P})}{\partial \mathbf{p}} \mathbf{B}^{\dagger}, \quad (63)$$

can be computed in a systematic and efficient manner using the recursive algorithm for the inverse dynamics computation of multibody systems (Ayusawa & Yoshida, 2018; Lee, Kim, Park, Kim, & Bobrow, 2005; Martin & Bobrow, 1999). We refer the reader to Lee (2019, p. 175–181) for detailed derivations and a pseudo-code description of the algorithm.

Gradients for the geometric trajectory excitation criterion proposed in this paper can be obtained by a slight modification of the above procedure. First, evaluating the cost is trivial via the following substitution for $\mathbf{M}_{\mathbf{B}}$:

$$\mathbf{M}_{\mathbf{B}} \leftarrow \mathbf{M}_{\mathbf{B}} \cdot \mathbf{H}_0^{-1}. \quad (64)$$

For the computation of the gradient, the following substitution is made in place of (61):

$$\frac{\partial \mathbf{M}_{\mathbf{B}}}{\partial \mathbf{p}} \leftarrow \frac{\partial \mathbf{M}_{\mathbf{B}}}{\partial \mathbf{p}} \cdot \mathbf{H}_0^{-1}, \quad (65)$$

since $\mathbf{H}_0$ is a given constant matrix.

5. Parameter identification for high-dimensional systems

For low-dimensional mechanical systems, e.g., standard industrial manipulators with seven or fewer dofs, obtaining sufficiently rich excitation trajectories and finding reliable estimates for the entire set of base parameters is more often than not feasible. This is generally not the case for more complex, high-dof structures like humanoids and multi-legged robots. Safety and other physical constraints (e.g., not falling over) can severely restrict the admissible dynamic motions (Bonnet, Pfeiffer, Fraisse, Crosnier, & Venture, 2018), and the available sensor measurements may be limited (e.g., not every joint may be equipped with a torque sensor, some parameters may be highly sensitive to sensor locations).

Attempting to find reference trajectories that sufficiently excite the entire set of parameters may therefore not be viable in practice. Based on the same geometric framework presented earlier, we propose a coordinate-invariant method for finding the optimal reduced set of parameters that can be reliably identified from a given set of feasible excitation trajectories. The key concept underlying our approach is to compare the respective sensitivities of the parameters based on the normalized covariance matrix.

The proposed reduced parameter identification strategy proceeds as follows:

- (1) Concatenate all the regressor matrices over the given feasible trajectories, and calculate the generalized inverse⁵ of the information matrix defined as follows:

$$\mathbf{C} = (\mathbf{M}_B)^g \triangleq \mathbf{H}_0^{-1/2} \left(\sum_{i=1}^{n_l} \lambda_i \mathbf{u}_i \mathbf{u}_i^T \right) \mathbf{H}_0^{-1/2}, \quad (66)$$

where eigen decomposition of the geometric normalized information matrix is given as

$$\mathbf{H}_0^{-1/2} \mathbf{M}_B \mathbf{H}_0^{-1/2} = \sum_{i=1}^{n_l} (1/\lambda_i) \mathbf{u}_i \mathbf{u}_i^T \quad (67)$$

with eigenvalues $0 < 1/\lambda_{n_l} \leq \dots \leq 1/\lambda_1$ ($n_l \leq N_B$) and corresponding eigenvectors $\mathbf{u}_i \in \mathbb{R}^{N_B}$.⁶

- (2) Transform $\mathbf{v}_i = \mathbf{H}_0^{-1/2} \mathbf{u}_i$, $\forall i = 1, \dots, n_l$, so that each $\mathbf{v}_i$ represents the direction of the i th smallest projected estimation variance of the maximum likelihood estimator characterized by the corresponding eigenvalue

$$\lambda_i = \text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle], \quad (68)$$

⁷ under the unit-norm and orthogonality constraints $\langle \mathbf{v}_i, \mathbf{v}_i \rangle = 1$ and $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ (which hold from $\mathbf{u}_i^T \mathbf{u}_i = 1$ and $\mathbf{u}_i^T \mathbf{u}_j = 0$). Also, define.

- (3) Set a parameter estimation variance threshold $\epsilon_\lambda > 0$ and define matrices

$$\mathbf{V}_- = [\mathbf{v}_1, \dots, \mathbf{v}_r], \quad (69)$$

$$\mathbf{\Lambda}_- = \text{diag}(\lambda_1, \dots, \lambda_r), \quad (70)$$

such that λ_r is the largest eigenvalue less than or equal to ϵ_λ .

- (4) Identify only the low estimation variance parameters by solving

$$\min_{\Phi_B \in \mathbb{R}^{N_B}, \mathbf{w} \in \mathbb{R}^r} \|\mathbf{A}_B \Phi_B - \mathbf{y}\|_{\Sigma^{-1}}^2 \quad (71)$$

$$\text{s.t. } \Phi_B = \Phi_B^0 + \mathbf{V}_- \cdot \mathbf{w}, \quad (72)$$

which has the following closed-form solution:

$$\begin{aligned} \Phi_B^* &= \Phi_B^0 + \mathbf{V}_- \cdot \mathbf{w}^* \\ \mathbf{w}^* &= \mathbf{\Lambda}_- \mathbf{V}_-^T \mathbf{A}_B^T \Sigma^{-1} (\mathbf{y} - \mathbf{A}_B \Phi_B^0) \end{aligned} \quad (73)$$

The parameter update on the reduced set of sufficiently excited parameters are given by $w_i^* = \langle \mathbf{v}_i, \Phi_B^* - \Phi_B^0 \rangle = \lambda_i \mathbf{v}_i^T \mathbf{A}_B^T \Sigma^{-1} (\mathbf{y} - \mathbf{A}_B \Phi_B^0)$, for $i = 1, \dots, r$.

The proposed method essentially updates only the reduced set of parameters with a certain level of confidence as specified by the eigenvalues λ_i , reflecting the sensitivities (or variances) of the projected parameters $w_i = \langle \mathbf{v}_i, \Phi_B \rangle$. For the updated set of parameters, the estimator is unbiased, i.e., $\mathbb{E}[\mathbf{w}^*] = \mathbf{w}$, with estimation variance,

$$\text{var}[\mathbf{w}^*] = \mathbf{\Lambda}_- \mathbf{V}_-^T \mathbf{A}_B^T \Sigma^{-1} \mathbf{A}_B \mathbf{V}_- \mathbf{\Lambda}_-, \quad (74)$$

and $\text{var}[w_i^*] = \lambda_i$ (c.f. (68)).

We point out that the presented parameter reduction strategy is not merely a combinatoric selection of the entries of the vector-valued parameters, but determines non-trivial linear combinations in a coordinate-free manner.

⁵ $\mathbf{X}^g \in \mathbb{R}^{n \times n}$ is generalized inverse of a (singular) matrix $\mathbf{X} \in \mathbb{R}^{n \times n}$, if it satisfies the condition $\mathbf{X} \mathbf{X}^g \mathbf{X} = \mathbf{X}$ (Rao & Mitra, 1972).

⁶ When $\mathbf{A}_B$ is full rank, $\mathbf{C}$ and $\mathbf{H}_0^{1/2} \mathbf{C} \mathbf{H}_0^{1/2}$ are simply the estimation covariance matrix and its geometric normalized counterpart for the base parameters.

⁷ From the property of generalized inverse of the information matrix (see Section 6.1 of Rao & Mitra, 1972), $\text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle] = (\mathbf{H}_0 \mathbf{v}_i)^T \mathbf{C} (\mathbf{H}_0 \mathbf{v}_i) = (\mathbf{H}_0^{1/2} \mathbf{u}_i)^T \mathbf{C} (\mathbf{H}_0^{1/2} \mathbf{u}_i) = \lambda_i$ holds.

Remark 8. Determining the scale of the estimation variance threshold value ϵ_λ in step 4 is eased by the fact that all the eigenvalues λ_i are coordinate-invariant, dimensionless values. Further, assuming that the prior distribution is given as $\Phi_B \sim \mathcal{N}(\Phi_B^0, \mathbf{H}_0^{-1}/\gamma)$, $\gamma \cdot \lambda_i$ indicates the relative estimation variance with respect to the prior variance projected along the direction $\mathbf{v}_i$:

$$\begin{aligned} \lambda_i &= \frac{\text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle]}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} = \frac{\text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle]}{\mathbf{v}_i^T \mathbf{H}_0 \mathbf{v}_i} \\ &= \frac{\text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle]}{(\mathbf{v}_i^T \mathbf{H}_0 \text{var}[\Phi_B]_{\text{prior}} \mathbf{H}_0 \mathbf{v}_i) \cdot \gamma} \\ &= \frac{\text{var}[\langle \mathbf{v}_i, \Phi_B^* \rangle]}{\text{var}[\langle \mathbf{v}_i, \Phi_B \rangle]_{\text{prior}}} \cdot \frac{1}{\gamma} \end{aligned} \quad (75)$$

Hence, in practice, the threshold value ϵ_λ should scale depending on the quality/confidence of the prior estimate.

Remark 9. The present framework can be extended to readily generate optimal excitation trajectories for the reduced set of effectively excitable parameters. That is, the parameters associated with directions of the largest estimation variances can be sequentially disregarded, until the optimized excitation criterion associated with the reduced set of parameters reduces to a reasonably small dimensionless value.

6. Experiments

Extensive numerical and hardware experiments involving a 2-dof SCARA robot, a 4-dof AMBIDEX cable-driven manipulator, and a 29-dof humanoid robot are presented. The main claims that we verify through our case studies are: (i) existing coordinate-dependent excitation criteria can be highly sensitive to the choice of coordinates; (ii) the proposed geometric excitation criterion more uniformly improves the identification accuracy of the typically poorly scaled parameters while guaranteeing coordinate-invariance; (iii) in cases where only poorly exciting trajectory samples are available, the proposed reduced identification framework can significantly improve the accuracy and robustness of identification even with fairly rough estimates of the prior parameter values.

6.1. SCARA robot

Numerical optimal excitation trajectory generation and identification experiment on a two-dof SCARA robot is conducted. Fig. 4(a) shows the kinematic specifications of the SCARA robot and unknown (uniform sphere) payload loaded at the end-effector link. Here we only excite and identify the inertial parameters of the payload attached to the second link, while assuming that the dynamics associated with the robot (without the payload) is known. The joint torque values are observed with additive Gaussian noise of standard deviation 0.03 Nm. The four base parameters associated with the payload are the mass m , x-y components of the first moment h^x, h^y , and rotational inertia about the z-axis I^{zz} .

Least-square parameter identification results are compared across the spectrum of pure unnormalized condition number criteria and their normalized counterparts for optimal excitation trajectory generation. Moreover, we examine the effects of using different body-fixed reference frames in adopting coordinate-dependent criterion that includes both the unnormalized and normalized criteria from Presse and Gautier (1993). Four body-fixed coordinate frames {1}, {2}, {3}, {4} are considered as shown in Fig. 4(a). Fifty joint kinematics samples, i.e., $\mathbf{P} = \{\mathbf{q}_i, \dot{\mathbf{q}}_i, \ddot{\mathbf{q}}_i\}_{i=1}^{50}$, used for identification are directly optimized using the SQP solver

Table 1
2-dof SCARA robot with unknown payload: identification results with “condition number” optimization.

Normalization	Coordinate (frame)	RMS error of the identified parameters (in coordinate {3})			
		m [kg]	$p^x = \frac{h^x}{m}$ [mm]	$p^y = \frac{h^y}{m}$ [mm]	$r^z = \sqrt{\frac{I^z}{m}}$ [mm]
None	1	0.03	0.78	1.27	5.24
	2	0.03	0.85	1.56	5.33
	3	0.03	0.70	1.40	5.14
	4	0.03	0.79	1.30	4.92
Diagonal (Presse & Gautier, 1993)	1	1.53	0.41	317.90	119.67
	2	0.04	1.07	1.39	2.87
	3	0.79	0.18	19.11	20.46
	4	0.07	2.31	2.19	4.15
Proposed	(Invariant)	0.06	0.60	1.09	1.56
Initial (unoptimized) traj.		0.04	0.16	149.82	89.02
Prior value Φ_B^0		4.58	0.02	1.94	86.07
Ground truth value		4.00	0.00	0.00	81.65

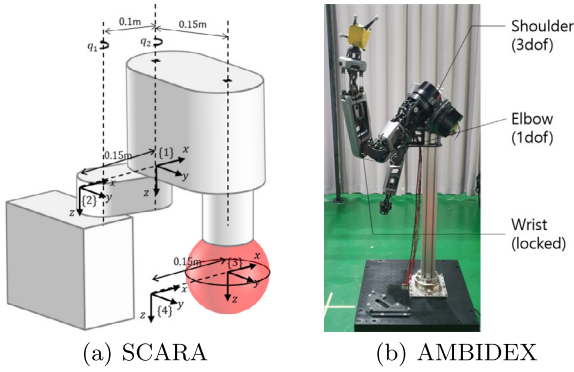


Fig. 4. The figures show 2-dof SCARA robot with unknown payload (red solid sphere) and reference frames considered, and joint/motor specifications of the AMBIDEX cable-driven manipulator.

of the MATLAB built-in function `fmincon` with recursive analytic gradient. The initial random excitation is set by uniformly sampling the values $q, \dot{q}, \ddot{q}$ within their pre-specified limits.

Table 1 compares the RMS errors of the identified parameters from using different coordinate frames and normalization methods for excitation trajectory generation. In order to quantify the individual parameter errors in a more physically intuitive way, the identified parameters of h^x, h^y, I^z are appropriately scaled with the identified mass m to represent the position of the center of mass p^x, p^y and length of the moment arm r^z . It can be observed that identification with the initial random samples results in significant errors for p^y and r^z . Such ill-conditioning of the parameter errors is alleviated to some extent by using the trajectory samples from the unnormalized condition number optimization. Meanwhile, the identification results from using the coordinate-dependent (diagonal) normalization (Presse & Gautier, 1993) vary considerably depending on the coordinate choices. In fact, the prior values of h_x evaluated in coordinate frames {1} and {3} turn out to be relatively small in magnitude compared to the other parameter values, leading to the normalization of the original condition number criterion in an undesirable way. In contrast, our proposed approach for normalization is completely free of the aforementioned downside arising from coordinate dependency, while also showing well-conditioned identification of the parameters.

6.2. AMBIDEX manipulator

In this section we describe hardware experiments on the AMBIDEX cable-driven manipulator (Choi et al., 2020; Kim, 2017).

AMBIDEX is a seven-dof robot manipulator with ten rigid body links bilaterally articulated by seven pairs of cables wired across the kinematically constrained bodies. Each cable is actuated by an electric motor. As the last three motors associated with the 3-dof low-inertia wrist mechanism exhibits particularly complex friction behavior, the wrist joint is kept locked in the current experiment. Consequently, we consider only the dynamics associated with five serially connected rigid bodies driven by four motors as shown in Fig. 4(b). The rigid body dynamic equations including the simple joint Coulomb friction model are given as

$$\tau = M(q, \Phi) \ddot{q} + b(q, \dot{q}, \Phi) + \text{diag}(\text{sgn}(\dot{q})) k_c \quad (76)$$

$$= \Gamma(q, \dot{q}, \ddot{q}) \Phi + \text{diag}(\text{sgn}(\dot{q})) k_c \quad (77)$$

$$= \Gamma_B(q, \dot{q}, \ddot{q}) \Phi_B + \text{diag}(\text{sgn}(\dot{q})) k_c \quad (78)$$

where $q \in \mathbb{R}^4$ is the motor angles, $\tau \in \mathbb{R}^4$ is the motor torques, $\Phi \in \mathbb{R}^{50}$ is the vector of full inertial parameters of the 5 bodies, and $k_c \in \mathbb{R}^4$ is the vector of joint Coulomb friction coefficients. We note that the total maximal identifiable parameters from the above can be characterized as the base parameters for the rigid body dynamics term, $\Phi_B = B\Phi \in \mathbb{R}^{31}$, and all the friction parameters k_c . We conduct excitation trajectory optimization with respect to only the inertial parameters Φ_B , as the joint Coulomb friction the magnitude of the regressor, $\text{sgn}(\dot{q}_i)$, is constant during any non-zero velocity interval. However, least squares parameter identification is performed on both the (base) inertial parameters and the friction parameters as follows:

$$\min_{\Phi_B, k_c} \|A_B \Phi_B + A_c k_c - y\|^2, \quad (79)$$

where A_B, A_c, y are respectively the concatenations of inertial regressor matrices Γ_B , friction regressor matrices and motor torque vectors over the excitation trajectory samples.

Excitation trajectories $q_i(t), i = 1, \dots, 4$ are parametrized with fifth order B-splines as follows:

$$q_i(t) = \sum_{j=1}^{n_f} p_{ij} B_{ij}(t) + B_{i0}(t), \quad 0 \leq t \leq T, \quad (80)$$

with $n_f = 10$ the number of control points and $B_{i0}(t)$ appropriately set to enforce boundary conditions: $q_i(0) = q_{i0}$, $q_i(T) = q_{iT0}$, and $\dot{q}(0), \dot{q}(T), \ddot{q}(0), \ddot{q}(T)$ all set to zero. The fixed duration of time T for the excitation is set to ten seconds, while samples are drawn uniformly at a 100 Hz sampling frequency. The optimization parameters are the coefficients of the basis functions, $B_{ij}(t)$, given by $P = \{p_{ij}\}_{i=1}^4 \}_{j=1}^{n_f}$. We use the SQP optimizer of the MATLAB built-in function `fmincon` with recursive analytic gradients to perform the parameter optimization of P . Motor angle and velocity limits are appropriately set as constraints to

Table 2
4-dof AMBIDEX manipulator: full identification results with “E-optimality” optimization.

Normalization	Coordinate (base $\mathbf{B}$)	RMS torque error [Nm]	
		Motor 1, 2	Motor 3, 4
None	$\mathbf{V}^T$	0.0087	0.0095
	$\mathbf{S} \cdot \mathbf{V}^T$	0.0551	0.0667
Diagonal (Presse & Gautier, 1993)	$\mathbf{S}_{\text{rand},1} \cdot \mathbf{V}^T$	0.0071	0.0079
	$\mathbf{S}_{\text{rand},2} \cdot \mathbf{V}^T$	0.0071	0.0066
Proposed	(Invariant)	0.0062	0.0059
Initial (unoptimized) traj.		0.0560	0.0685
Prior value Φ_B^0 (from CAD)		0.0333	0.0186

the solver. The initial guess provided to the solver is chosen by randomly sampling the parameters $\mathbf{P}$ until the motor angle and velocity limit constraints are satisfied.

E-optimality is used as the excitation criterion. Both the unnormalized E-optimal criterion and its normalized counterparts are considered as in the previous experiment. The prior inertial parameter values of each link used for the normalization of the excitation criteria are obtained from the CAD model. We also examine the effects of using different coordinate choices for the base parameter representation in adopting the existing coordinate-dependent excitation criteria. However, unlike in the case of the previous experiment where different coordinate choices for the base parameters are made by selecting different sets of reference frames attached to the links, we now set multiple base parameter representations by applying different non-singular linear transformations to the reference base parameter representation obtained using the numerical SVD-based technique proposed in Atkeson et al. (1986). To explain, we first perform numerical SVD on the full regressor matrix $\mathbf{A}^{\text{init}}$ concatenated over the initial excitation trajectory samples, i.e., $\mathbf{A}^{\text{init}} = \mathbf{U}\mathbf{S}\mathbf{V}^T$. Then two different choices of base parameter representations, $\mathbf{B} = \mathbf{V}^T$ and $\mathbf{B} = \mathbf{S} \cdot \mathbf{V}^T$, are adopted in the unnormalized E-optimal criterion. For the normalized E-optimal criterion proposed in Presse and Gautier (1993), two other choices, $\mathbf{B} = \mathbf{S}_{\text{rand},1} \cdot \mathbf{V}^T$ and $\mathbf{B} = \mathbf{S}_{\text{rand},2} \cdot \mathbf{V}^T$, are adopted where $\mathbf{S}_{\text{rand},1}$ and $\mathbf{S}_{\text{rand},2}$ are the two randomly sampled non-singular square matrices.

Each E-optimal excitation trajectory is repeatedly executed on the hardware for six times, and parameter identification is performed separately using the samples obtained from each tryout. Since the ground truth model parameter values are not available, the accuracy of the identified parameters are examined based on the motor torque prediction error evaluated with respect to the validation data set. For fair comparison, validation data set is collected by executing D-optimal excitation trajectory with duration of 20 s. Note again that D-optimality criterion is always invariant to the coordinate choices regardless of the normalization.

Table 2 shows, for each of the different E-optimality criteria, the mean value of the RMS torque prediction errors (evaluated over the six set of repeatedly identified parameters). It can be observed that the excitation trajectory from the proposed normalized criteria come in the best parameter accuracy, to highlight again, with an added advantage of guaranteed invariance to the coordinate choices. Meanwhile, it is perhaps noticeable that the accuracy of the parameters identified using the unnormalized E-optimal excitation trajectory samples with the base parameter choice, $\mathbf{B} = \mathbf{S}\mathbf{V}^T$, show little difference compared to the case of using the initial trajectory samples for identification. In fact, the local gradient of the corresponding E-optimal criterion turns out to be near zero at the initial guess from which the SQP solver failed to converge to a sufficiently exciting trajectory.

We also examine how the reduced identification strategy provided in Section 5 can improve the parameter identification

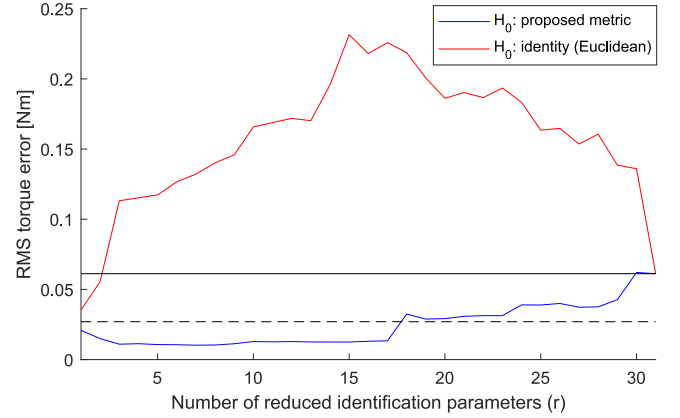


Fig. 5. Plot of RMS torque prediction error while varying the number r of the reduced identification parameters. The bold black horizontal line indicates the RMS torque error from the full identification, and the dashed black horizontal line indicates the RMS torque error from fixing all the base inertial parameters with the prior values, i.e., $\Phi_B = \Phi_B^0$.

accuracy given the poorly exciting trajectory samples.⁸ Fig. 5 shows the reduced identification result using the unnormalized E-optimal excitation trajectory computed under the base parameter choice, $\mathbf{B} = \mathbf{S} \cdot \mathbf{V}^T$. It can be observed that reducing the number of parameters for identification significantly improves the torque prediction performance compared to the previous full identification result; the lowest validation error of 0.010 Nm (achieved around $r = 5 \sim 10$) is more than 80% reduction compared to the full identification result. Note however that the performance improvement gained through the parameter reduction would in general be limited by the constant bias present in the prior estimate. Perhaps, what appears to be more interesting in Fig. 5 is that the same reduced identification method, but using the unnormalized (conventional) estimation covariance matrix, i.e., replacing the metric $\mathbf{H}_0$ with identity matrix, rather degrades the parameter identification accuracy. For all possible sizes of identified parameter combinations, the resulting validation errors are either worse than the full identification result or the one by fixing all the base inertial parameters with the prior values, mostly by large margins. This clearly demonstrates the importance of coordinate invariance in quantifying the parameter estimation variance or equivalently the information on the excitation trajectory samples.

6.3. Humanoid robot

In this section, we further demonstrate the efficacy of our reduced identification framework with a numerical experiment on

⁸ For the reduced identification, friction parameters are identified together with the reduced set of inertial parameters by simply replacing (71) with (79) in the original formulation

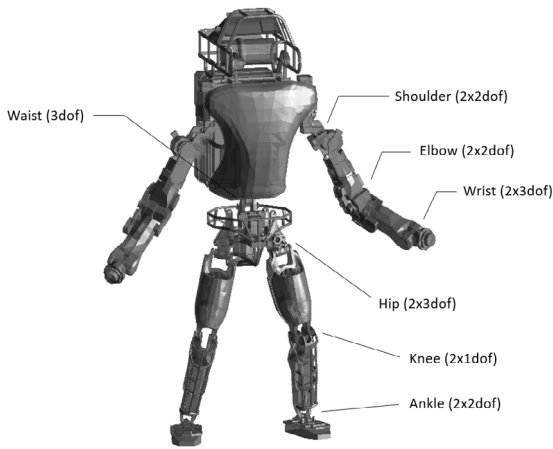


Fig. 6. Atlas V5 joint specifications.

a high-dimensional humanoid robot. Atlas V5 humanoid model having 29 joints and 30 body links is considered (See Fig. 6). The admissible range of whole body motion of the humanoid is restricted to be in dynamic balance with a single foot placed onto the flat ground. Rather than having direct observations on the joint torques, we adopt a single standard force plate placed under the support foot which measures 6-axis external Ground Reaction (GR) force and moment (Ayusawa et al., 2014). We note that the base parameters associated with the GR force/moment observations always sufficiently characterize the base parameters associated with the joint space dynamics of the open-chain structure with base fixed to the ground. Readers may refer to Raucourt, Campion, Bastin, Samin, and Willems (1991) for more details on the GR force/moment based robot model identification formulation. The force and moment outputs are observed with additive zero-mean Gaussian noise of standard deviations 0.5 N and 1.5 Nm respectively.

The proposed reduced identification strategy is tested upon a set of randomly generated dynamically feasible trajectories. The random trajectories are sampled by, first, sampling the feasible static initial and final joint configurations; the positions of the center of mass of the whole body projected to the ground are checked if they lie inside the region of contact between the foot and ground. Then each pair of these boundary configurations are connected with B-spline trajectory, whose control points are repeatedly random-sampled until the dynamic balance constraints as well as joint angle, velocity limit constraints are satisfied at the time sampling frequency of 15 Hz. Out of 20 random generated feasible trajectories each with time duration of 3 s (Fig. 7), 10 motions are used for identification and the remaining 10 motions are used for validation. The trajectory samples are drawn at the sampling frequency of 300 Hz. True inertial parameter values are determined from CAD data, and prior values are set by deliberately perturbing the true ones by adding random point masses to each of the bodies as similarly done in Lee et al. (2019) (the scale of perturbation for each of the bodies is set proportional to the true mass of the body). The scale-free variance threshold value ϵ_λ for determining the size r of the reduced set of parameters is set as $\epsilon_\lambda = 10^{-2}$.

Referring to Fig. 8, 70 number of sufficiently excited parameters out of 204 base parameters are selected and identified using the proposed reduced identification strategy. We would like to highlight that, although the GR force/moment observations are used to identify the parameters, the same source of observations need not only be considered for validating the identified parameters. In the current experiment, we examine how well the

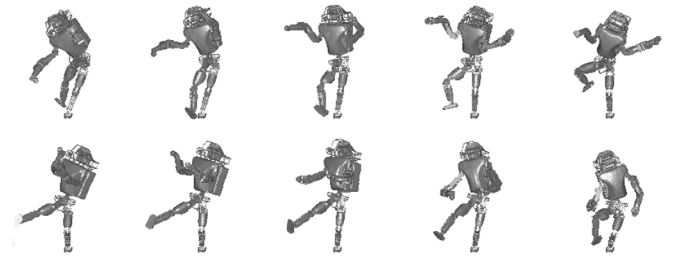


Fig. 7. Snapshots of 10th (top) and 20th (bottom) balancing motion trajectories used for (reduced) dynamic identification and validation.

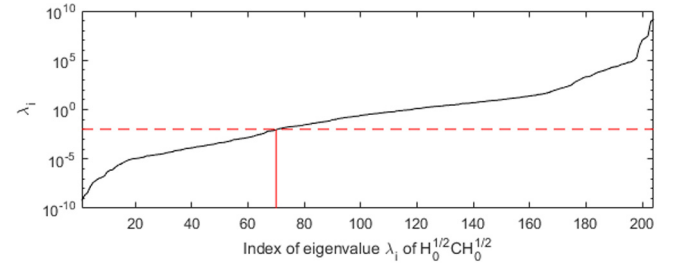


Fig. 8. Plot of eigenvalues of the normalized covariance matrix $H_0^{1/2} C H_0^{1/2}$ for Atlas V5 balancing trajectories. The horizontal dashed line represents the threshold value $\epsilon_\lambda = 10^{-2}$ used for parameter selection.

parameters identified from the GR force/moment observations can be generalized to match with the joint torque observations as well as the GR force/moment observations.

Tables 3 and 4 show respectively the RMS GR force/moment prediction error and RMS joint torque prediction error on the validation trajectory samples. It can be clearly observed that the parameter estimates from the reduced identification generalizes significantly better in terms of both the GR force/moment and the joint torques. It is perhaps remarkable that the prior inertial parameter values used for the reduced identification are very rough estimates of the true inertial parameters, as they result in significantly large output prediction errors compared to both the full and reduced identification results. Note again that out of 204 number of erroneous prior base parameters only 70 number of linear combinations are updated in the reduced identification.

7. Conclusion

We have presented a coordinate-invariant framework of optimal excitation trajectory generation for multibody mechanical systems identification. Practical benefit of using the proposed normalized criterion for optimal excitation lies not only in achieving improved robustness of parameter identification, but also in providing provably coordinate-invariant designs. The proposed normalization of covariance matrix is also shown to be useful for assessing the effective identifiability of the parameters and conducting reduced identification for complex high-dimensional mechanical systems like humanoids.

Although we have focused only on the inertial parameters of coupled rigid body systems, we note that other important mechanical properties, including stiffness/impedance and frictions, should also be treated in a similar geometric manner. Scale invariance seems enough for, e.g., joint stiffness or friction parameters, whereas matrix-valued spatial stiffness/impedance or damping exhibit a change of coordinate also in a form of affine transformations. Applying our coordinate-invariant framework of parameter excitation/identification to these more general types of mechanical systems is a direction for future work. Finally,

Table 3

Identification results on Atlas V5 humanoid with parameter reduction: direct validation on GR force/moment.

	GR force [N]	GR moment [Nm]
Prior	61.768	11.770
Full ID.	0.017	0.141
Reduced ID.	0.009	0.024

Table 4

Identification results on Atlas V5 humanoid with parameter reduction: validation on the joint torques.

	RMS torque error $\tau - \hat{\tau}$ [Nm]						
	Hip (2 × 3dof)	Knee (2 × 1dof)	Ankle (2 × 2dof)	Waist (3dof)	Shoulder (2 × 2dof)	Elbow (2 × 2dof)	Wrist (2 × 3dof)
Prior	11.279	14.834	10.101	11.717	1.011	0.304	0.062
Full ID.	0.131	0.132	0.119	0.135	0.053	0.034	0.015
Reduced ID.	0.019	0.017	0.034	0.018	0.009	0.008	0.002

the presented geometric framework in the context of general experimental design possess a clear limitation in that the linear observation model is assumed. Extending our approach to nonlinear models, in which extended class of observations and model parameters can be taken into account, appears to be another interesting direction of future research

Appendix. Proof of Proposition 5

We first prove that the alternate definition (49) of our criterion is invariant to the coordinate choices, and then prove it is in fact equivalent to the original definition (48) which explicitly involves arbitrary choice of base parameter representation through the decomposition $\mathbf{A} = \mathbf{A}_B \mathbf{B}$.

Denote by Φ_a , $\mathbf{A}_a$, and Σ_a the inertial parameters, regressor matrix, and observation covariance matrix, respectively, evaluated under some particular choice of coordinate; the superscript a represents the coordinate choice. First, recall from (35) that under the change of coordinate from b to a , the pseudo-inertia matrix representation of inertial parameters always admits the linear transformation rule, say $\Psi_{ab} : \mathcal{P}(4) \rightarrow \mathcal{P}(4)$ of the form, $\mathbf{P}_a = \Psi_{ab}(\mathbf{P}_b) \triangleq \mathbf{G}_{ab} * \mathbf{P}_b = \mathbf{G}_{ab} \mathbf{P}_b \mathbf{G}_{ab}^T$, for some constant nonsingular square matrix $\mathbf{G}_{ab} \in \mathbb{R}^{4 \times 4}$. By the linearity of the mappings $\mathbf{P}$ and Ψ_{ab} , the vector representation of inertial parameters ϕ hence always admits the linear transformation rule

$$\psi_{ab} = \mathbf{P} \circ \Psi_{ab} \circ \mathbf{P}^{-1} : \mathbb{R}^{10} \rightarrow \mathbb{R}^{10}, \quad (\text{A.1})$$

i.e., $\phi_a = \psi_{ab}(\phi_b) = \mathbf{P}^{-1}(\Psi_{ab}(\mathbf{P}(\phi_b)))$. Therefore, the full set of inertial parameters Φ (which is simply the concatenation of the inertial parameters ϕ_i of all the bodies) also admits a linear coordinate transformation rule, i.e., $\Phi_a = \mathbf{S}_{ab} \Phi_b$ for some constant nonsingular square matrix $\mathbf{S}_{ab}$. Now, recalling the definition of the Fisher information matrix, Fisher information matrix $\mathbf{M}$ on the full set of inertial parameter Φ is given by,

$$\mathbf{M} \triangleq \mathbb{E} [\nabla_{\Phi} \log p(\mathbf{y}|\Phi) \cdot \nabla_{\Phi} \log p(\mathbf{y}|\Phi)^T] \quad (\text{A.2})$$

$$= \mathbf{A}^T \Sigma^{-1} \mathbf{A}. \quad (\text{A.3})$$

Using the chain rule, $\nabla_{\Phi_b} \log p(\mathbf{y}_b|\Phi_b) = \mathbf{S}_{ab}^T \nabla_{\Phi_a} \log p(\mathbf{y}_a|\Phi_a)$, coordinate transformation on the Fisher information matrix $\mathbf{M}$ is given by, $\mathbf{M}_a = \mathbf{S}_{ab}^T \mathbf{M}_b \mathbf{S}_{ab}^{-1}$, and hence,

$$\mathbf{A}_a^T \Sigma_a^{-1} \mathbf{A}_a = \mathbf{S}_{ab}^{-T} (\mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b) \mathbf{S}_{ab}^{-1}. \quad (\text{A.4})$$

From the coordinate invariance of the metric, i.e.,

$$ds^2 = d\Phi_{0a}^T \mathbf{G}_{0a} d\Phi_{0a} = d\Phi_{0b}^T \mathbf{S}_{ab}^T \mathbf{G}_{0a} \mathbf{S}_{ab} d\Phi_{0b}, \quad (\text{A.5})$$

it follows that

$$\mathbf{G}_{0a} = \mathbf{S}_{ab}^{-T} \mathbf{G}_{0b} \mathbf{S}_{ab}^{-1}. \quad (\text{A.6})$$

From (A.4) and (A.6), the transformation rule for $\mathbf{D} = \mathbf{A}^T \Sigma^{-1} \mathbf{A} \mathbf{G}_0^{-1}$ is then given by

$$\mathbf{D}_a = \mathbf{S}_{ab}^{-T} \mathbf{D}_b \mathbf{S}_{ab}^T. \quad (\text{A.7})$$

Since the set of eigenvalues of a matrix $\mathbf{X}$ is always preserved under the transformation $\mathbf{X} \mapsto \mathbf{G}^{-1} \mathbf{X} \mathbf{G}$ for any nonsingular matrix $\mathbf{G}$, we conclude that $f(\lambda_{\neq 0}(\mathbf{D}))$ is coordinate invariant.

We now show that the nonzero eigenvalues of $\mathbf{A}^T \Sigma^{-1} \mathbf{A} \mathbf{G}_0^{-1}$ are identical to the eigenvalues of $\mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b \mathbf{H}_0^{-1}$ regardless of the choice of $\mathbf{B}$, which suffices to complete the proof. The characteristic equation for $\mathbf{A}^T \Sigma^{-1} \mathbf{A} \mathbf{G}_0^{-1}$ is given by

$$\begin{aligned} p(s) &= \det(s\mathbf{I} - \mathbf{A}^T \Sigma^{-1} \mathbf{A} \mathbf{G}_0^{-1}) \\ &= \det(s\mathbf{I} - \mathbf{B}^T \mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b \mathbf{B} \mathbf{G}_0^{-1}) \\ &= s^{10-n_B} \cdot \det(s\mathbf{I} - \mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b \mathbf{B} \mathbf{G}_0^{-1} \mathbf{B}^T) \\ &= s^{10-n_B} \cdot \det(s\mathbf{I} - \mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b \mathbf{H}_0^{-1}), \end{aligned} \quad (\text{A.8})$$

where the last equality holds by Sylvester's determinant identity, i.e., $\det(\mathbf{I} + \mathbf{X} \mathbf{Y}) = \det(\mathbf{I} + \mathbf{Y} \mathbf{X})$ for square matrices $\mathbf{X} \mathbf{Y}$ and $\mathbf{Y} \mathbf{X}$. Therefore, n_B number of nonzero eigenvalues of $\mathbf{A}^T \Sigma^{-1} \mathbf{A} \mathbf{G}_0^{-1}$ exactly match the positive eigenvalues of $\mathbf{A}_b^T \Sigma_b^{-1} \mathbf{A}_b \mathbf{H}_0^{-1}$. $\square$

References

- Armstrong, Brian (1989). On finding exciting trajectories for identification experiments involving systems with nonlinear dynamics. *International Journal of Robotics Research*, 8(6), 28–48.
- Atkeson, Christopher G., An, Chae H., & Hollerbach, John M. (1986). Estimation of inertial parameters of manipulator loads and links. *International Journal of Robotics Research*, 5(3), 101–119.
- Ayusawa, Ko, Rioux, Antoine, Yoshida, Eiichi, Venture, Gentiane, & Gautier, Maxime (2017). Generating persistently exciting trajectory based on condition number optimization. In *Robotics and automation, 2017 IEEE international conference on* (pp. 6518–6524). IEEE.
- Ayusawa, Ko, Venture, Gentiane, & Nakamura, Yoshihiko (2014). Identifiability and identification of inertial parameters using the underactuated base-link dynamics for legged multibody systems. *International Journal of Robotics Research*, 33(3), 446–468.
- Ayusawa, Ko, & Yoshida, Eiichi (2018). Comprehensive theory of differential kinematics and dynamics towards extensive motion optimization framework. *International Journal of Robotics Research*, Article 027836491877289.
- Bonnet, Vincent, Fraisse, Philippe, Crosnier, André, Gautier, Maxime, González, Alejandro, & Venture, Gentiane (2016). Optimal exciting dance for identifying inertial parameters of an anthropomorphic structure. *IEEE Transactions on Robotics*, 32(4), 823–836.
- Bonnet, Vincent, Pfeiffer, Kai, Fraisse, Philippe, Crosnier, André, & Venture, Gentiane (2018). Self-generation of optimal exciting motions for identification of a humanoid robot. *International Journal of Humanoid Robotics*, 15(6), Article 1850024.
- Chaloner, Kathryn (1984). Optimal Bayesian experimental design for linear models. *The Annals of Statistics*, 12(1), 283–300.
- Choi, Keunjun, Kwon, Jaewoon, Lee, Taeyoon, Park, Changwoo, Pyo, Jinwon, Lee, Choongin, et al. (2020). A hybrid dynamic model for the AMBIDEX tendon-driven manipulator. *Mechatronics*, 69, Article 102398.

- Detle, Holger (1997). Designing experiments with respect to 'standardized' optimality criteria. *Journal of the Royal Statistical Society. Series B. Statistical Methodology*, 59(1), 97–110.
- Donev, A. N., Atkinson, A. C., & Tobias, R. D. (2007). *Oxford statistical science series, Optimum experimental designs, with SAS*. United Kingdom: Oxford University Press.
- Eells, James, & Sampson, Joseph H. (1964). Harmonic mappings of Riemannian manifolds. *American Journal of Mathematics*, 86(1), 109–160.
- Gautier, M., Janot, A., & Vandanjon, P. O. (2008). DIDIM: A new method for the dynamic identification of robots from only torque data. In *2008 IEEE international conference on robotics and automation* (pp. 2122–2127).
- Gautier, Maxime, & Khalil, Wisama (1992). Exciting trajectories for the identification of base inertial parameters of robots. *International Journal of Robotics Research*, 11(4), 362–375.
- Gevers, Michel (2005). Identification for control: From the early achievements to the revival of experiment design*. *European Journal of Control*, 11(4), 335–352.
- Janot, A., Vandanjon, P. O., & Gautier, M. (2009). Using robust regressions and residual analysis to verify the reliability of LS estimation: Application in robotics. In *2009 IEEE/RSJ international conference on intelligent robots and systems* (pp. 1962–1967).
- Janot, A., Vandanjon, P., & Gautier, M. (2014). A generic instrumental variable approach for industrial robot identification. *IEEE Transactions on Control Systems Technology*, 22(1), 132–145.
- Jansson, Henrik, & Hjalmarsson, Håkan (2005). Optimal experiment design in closed loop. *IFAC Proceedings Volumes*, 38(1), 488–493, 16th IFAC World Congress.
- Jazar, Reza N. (2007). *Theory of applied robotics: Kinematics, dynamics, and control*. Springer Publishing Company, Incorporated.
- Kim, Yong-Jae (2017). Anthropomorphic low-inertia high-stiffness manipulator for high-speed safe interaction. *IEEE Transactions on Robotics*, 33(6), 1358–1374.
- Lee, Taeyoon (2019). *Geometric methods for dynamic model-based identification and control of multibody systems*. Seoul National University.
- Lee, Sung-Hee, Kim, Junggon, Park, F. C., Kim, Munsang, & Bobrow, J. E. (2005). Newton-type algorithms for dynamics-based robot movement optimization. *IEEE Transactions on Robotics*, 21(4), 657–667.
- Lee, Taeyoon, Kwon, Jaewoon, & Park, Frank C. (2018). A natural adaptive control law for robot manipulators. In *2018 IEEE/RSJ international conference on intelligent robots and systems*.
- Lee, Taeyoon, & Park, Frank C. (2018). A geometric algorithm for robust multibody inertial parameter identification. *IEEE Robotics and Automation Letters*, 3(3), 2455–2462.
- Lee, Taeyoon, Wensing, Patrick M., & Park, Frank C. (2019). Geometric robot dynamic identification: A convex programming approach. *IEEE Transactions on Robotics*, 36(2), 348–365.
- López-Fidalgo, J., Rivas-López, M. J., & Fernández-Garzón, B. (2007). A-optimality standardized through the coefficient of variation. *Communications in Statistics. Theory and Methods*, 36(4), 781–792.
- Martin, Bryan J., & Bobrow, James E. (1999). Minimum-effort motions for open-chain manipulators with task-dependent end-effector constraints. *International Journal of Robotics Research*, 18(2), 213–224.
- Moakher, Maher, & Batchelor, Philipp G. (2006). Symmetric positive-definite matrices: From geometry to applications and visualization. In *Visualization and processing of tensor fields* (pp. 285–298). Springer.
- Morgan, John P., & Wang, Xiaowei (2010). Weighted optimality in designed experimentation. *Journal of the American Statistical Association*, 105(492), 1566–1580.
- Olsen, Martin M., Swevers, Jan, & Verdonck, Walter (2002). Maximum likelihood identification of a dynamic robot model: Implementation issues. *International Journal of Robotics Research*, 21(2), 89–96.
- Presse, C., & Gautier, Maxime (1993). New criteria of exciting trajectories for robot identification. In *Robotics and automation, 1993. Proceedings., 1993 IEEE international conference on* (pp. 907–912). IEEE.
- Pronzato, Luc (2008). Optimal experimental design and some related control problems. *Automatica*, 44(2), 303–325.
- Pukelsheim, Friedrich (2006). *Optimal design of experiments (classics in applied mathematics, 50)*. Philadelphia, PA, USA: Society for Industrial and Applied Mathematics.
- Ramdan, N., & Poignet, P. (2005). Robust dynamic experimental identification of robots with set membership uncertainty. *IEEE/ASME Transactions on Mechatronics*, 10(2), 253–256.
- Rao, C. Radhakrishna, & Mitra, Sujit Kumar (1972). Generalized inverse of a matrix and its applications. In *Proceedings of the sixth Berkeley symposium on mathematical statistics and probability, volume 1: Theory of statistics* (pp. 601–620). Berkeley, Calif.: University of California Press.
- Raucourt, B., Campion, G., Bastin, G., Samin, J. C., & Willems, P. Y. (1991). On the identification of the barycentric parameters of robot manipulators from external measurements. In *Proceedings. 1991 IEEE international conference on robotics and automation*, vol. 2 (pp. 1169–1174). <http://dx.doi.org/10.1109/ROBOT.1991.131767>.
- Rojas, Cristian R., Welsh, James S., Goodwin, Graham C., & Feuer, Arie (2007). Robust optimal experiment design for system identification. *Automatica*, 43(6), 993–1008.
- Schwabe, Rainer (2012). *Optimum designs for multi-factor models, volume 113*. Springer Science & Business Media.
- Sousa, Cristóvão D., & Cortesão, Rui (2014). Physical feasibility of robot base inertial parameter identification: A linear matrix inequality approach. *International Journal of Robotics Research*, 33(6), 931–944.
- Swevers, Jan, Ganseman, Chris, Bilgin, Dilek, De Schutter, Joris, & Van Brussel, Hendrik (1997). Optimal robot excitation and identification. *IEEE Transactions on Robotics and Automation*, 13(5), 730–740.
- Venture, Gentiane, Ripert, P. -J., Khalil, Wisama, Gautier, Maxime, & Bodson, Philippe (2006). Modeling and identification of passenger car dynamics using robotics formalism. *IEEE Transactions on Intelligent Transportation Systems*, 7(3), 349–359.
- Wensing, Patrick M., Kim, Sangbae, & Slotine, Jean-Jacques E. (2018). Linear matrix inequalities for physically consistent inertial parameter identification: A statistical perspective on the mass distribution. *IEEE Robotics and Automation Letters*, 3(1), 60–67.
- Wensing, Patrick M., Niemeyer, Günter, & Slotine, Jean-Jacques E. (2017). Observability in inertial parameter identification. *CoRR*, abs/1711.03896.
- Wilson, Andrew D., Schultz, Jarvis A., & Murphey, Todd D. (2014). Trajectory synthesis for Fisher information maximization. *IEEE Transactions on Robotics*, 30(6), 1358–1370.



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